UP Board Class 12 Biology - 348 (CM) 2023 Question Paper with Solutions

Time Allowed :3 Hours	**Maximum Marks :**100	**Total Questions :**9

General Instructions

Read the following instructions very carefully and strictly follow them:

1. There are in all nine questions in this question paper.
2. All questions are compulsory.
3. In the beginning of each question, the number of parts to be attempted are clearly mentioned.
4. Marks allotted to the questions are indicated against them.
5. Start solving from the first question and proceed to solve till the last one. Do not waste your time over a question you cannot solve.

Multiple Choice Questions

1.(a) Which one is found in pollen grain?

(A) Two male gametes
(B) Two vegetative cells and one generative cell
(C) One generative cell and one tube cell
(D) One tube cell and one male gamete

Correct Answer: (C) One generative cell and one tube cell

Solution:
Step 1: Structure of pollen grain.
A mature pollen grain in angiosperms is typically two-celled, consisting of: 1. One large vegetative (tube) cell.
2. One smaller generative cell.

Step 2: Role of the cells.
- The tube cell elongates to form the pollen tube during fertilization.
- The generative cell undergoes mitotic division to produce two male gametes that participate in double fertilization.

Step 3: Analysis of options.
- (A) Two male gametes: Present only after the generative cell divides; not initially in the

pollen grain.
- (B) Two vegetative cells and one generative cell: Incorrect; only one vegetative cell is present.
- (C) One generative cell and one tube cell: Correct; this is the typical structure of a mature pollen grain.
- (D) One tube cell and one male gamete: Incorrect; the male gametes form after generative cell division.

Step 4: Conclusion.
The correct answer is (C) One generative cell and one tube cell.

Quick Tip

A pollen grain is typically two-celled: one tube cell and one generative cell. The generative cell later forms two male gametes for double fertilization.

(b) Number of pollen sacs in an anther is:

(A) Four
(B) Two
(C) One
(D) Eight

Correct Answer: (A) Four

Solution:
Step 1: Structure of an anther.
In angiosperms, a typical anther is bilobed, and each lobe contains two pollen sacs. Therefore, a single anther consists of four pollen sacs in total.

Step 2: Function of pollen sacs.
Pollen sacs serve as the site for microsporogenesis, where microspore mother cells undergo meiosis to form haploid microspores, which develop into pollen grains.

Step 3: Analysis of options.
- (A) Four: Correct, as a standard anther contains four pollen sacs.
- (B) Two: Incorrect, this refers only to one lobe of the anther.
- (C) One: Incorrect, since anthers always have more than one sac.
- (D) Eight: Incorrect, as the number exceeds the typical count.

Step 4: Conclusion.
The correct answer is (A) Four, as a typical anther has four pollen sacs.

Quick Tip

A bilobed anther has two lobes, and each lobe contains two pollen sacs, making a total of four pollen sacs per anther.

(c) How many meiotic divisions are required to get 200 grains of Barley?

(A) 200
(B) 100
(C) 250
(D) 50

Correct Answer: (B) 100

Solution:
Step 1: Understanding the requirement.
Each grain of barley is essentially a seed, which originates from a single fertilized embryo sac. The formation of one functional embryo sac requires one meiotic division in the megaspore mother cell.

Step 2: Considering double fertilization.
In angiosperms, each embryo sac is formed as a result of a single meiotic event. Although many pollen grains are produced, for this calculation we focus only on ovule development, i.e., seed (grain) formation.

Step 3: Calculation.
- One meiotic division in the megaspore mother cell produces four megaspores, but only one becomes functional.
- Thus, one meiotic division results in the formation of one embryo sac (and eventually one seed).
- Therefore, to form 200 grains of barley:

$$200 \div 2 = 100 \,\text{meiotic divisions}$$

(because only one out of four megaspores formed per meiosis is functional).

Step 4: Conclusion.
The correct answer is (B) 100 meiotic divisions for producing 200 grains of barley.

Quick Tip

In angiosperms, only one functional megaspore is produced per meiosis, hence the number of meiotic divisions equals half the number of seeds formed.

(d) Congenital metabolic disorder is:

(A) Phenylketonuria
(B) Colour blindness
(C) Haemophilia
(D) Anaemia

Correct Answer: (A) Phenylketonuria

Solution:
Step 1: Understanding congenital metabolic disorders.
Congenital metabolic disorders are inherited genetic conditions that impair enzyme functions or biochemical pathways from birth, leading to abnormal metabolism of specific substances.

Step 2: Analysis of options.
- (A) Phenylketonuria: This is a congenital metabolic disorder caused by a mutation resulting in deficiency of the enzyme phenylalanine hydroxylase. As a result, phenylalanine cannot be metabolized properly, leading to its accumulation and severe neurological damage.
- (B) Colour blindness: Although genetic, it is a visual disorder caused by defects in cone cells of the retina, not a metabolic disorder.
- (C) Haemophilia: A hereditary bleeding disorder due to deficiency of clotting factors, but not related to metabolism.
- (D) Anaemia: Typically caused by nutritional deficiencies (e.g., iron) or other health conditions, and is not classified as a congenital metabolic disorder.

Step 3: Conclusion.
The correct answer is (A) Phenylketonuria, as it is a congenital metabolic disorder.

Quick Tip

Phenylketonuria (PKU) is an inborn error of metabolism caused by the absence of phenylalanine hydroxylase enzyme.

Subjective Questions

(a) Mention the name of causal organism of Typhoid fever.

Solution:
Name of the Organism:

The causal organism of **Typhoid fever** is ***Salmonella typhi***, a pathogenic bacterium belonging to the family Enterobacteriaceae.

Characteristics of *Salmonella typhi*:

- It is a rod-shaped, Gram-negative bacterium.
- It is motile with flagella.
- It is a facultative anaerobe (can live with or without oxygen).

Mode of Transmission:
- Spread mainly through contaminated food, milk, and water.
- Carriers such as flies may transmit the pathogen.
- Human carriers (recovered patients who still harbor bacteria in the gall bladder) can also spread the disease.

Symptoms of Typhoid Fever:

1. Sustained high fever (39°–40°C).
2. Weakness, headache, abdominal pain, constipation or diarrhea.
3. Enlarged spleen and liver.
4. Appearance of rose-colored spots on the chest and abdomen in some patients.

Diagnosis:
- Confirmed by **Widal test**, which detects antibodies against *Salmonella typhi*.

Prevention and Control:
- Proper sanitation and personal hygiene.
- Use of clean drinking water and safe food practices.
- Vaccination against typhoid (Ty21a, Vi polysaccharide vaccines).

Conclusion:
Typhoid fever is a water-borne disease caused by *Salmonella typhi*. Preventive measures such as hygiene, sanitation, and vaccination are highly effective in controlling its spread.

> Quick Tip
>
> Typhoid fever is caused by *Salmonella typhi*, spreads via contaminated food and water, diagnosed by Widal test, and prevented by vaccination and sanitation.

(b) What type of nucleus is found in bacterial cell?

Solution:
Type of Nucleus: Bacteria possess a **prokaryotic nucleus**, also called a **nucleoid**. It differs significantly from the eukaryotic nucleus found in higher organisms.
Structure of Bacterial Nucleoid:

- It is not surrounded by a nuclear membrane.
- DNA is present as a single, circular, double-stranded molecule.
- The DNA is "naked" – not associated with histone proteins.
- No nucleolus is present.

Additional Genetic Material: - Many bacteria also possess small circular DNA fragments called **plasmids**, which carry genes for antibiotic resistance and other traits.
Functions of Nucleoid:

1. Controls cell metabolism and division.
2. Stores genetic information and passes it to daughter cells during binary fission.
3. Regulates synthesis of proteins by transcription and translation.

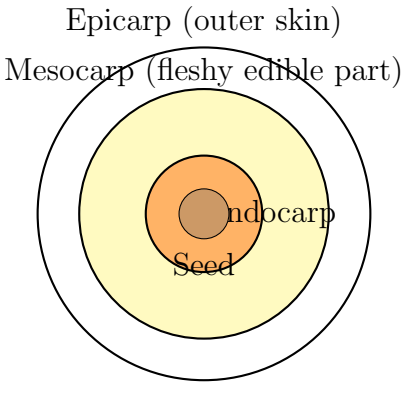
Conclusion: Thus, bacterial cells have a primitive nucleus called **nucleoid**, which lacks a nuclear membrane, histones, and nucleolus. It is a defining feature of prokaryotes.

> Quick Tip
>
> Bacteria = Prokaryotes → Nucleoid with naked circular DNA (no nuclear membrane, no histones).

(c) Draw a labelled diagram of a succulent fruit.

Solution:

A succulent fruit is a fleshy fruit in which the pericarp becomes soft, juicy, and edible. Examples include mango, tomato, guava, and orange.
Diagram:



> Quick Tip
>
> Succulent fruits have three layers: epicarp (skin), mesocarp (fleshy part), and endocarp (encloses seed).

(d) Describe in brief about advantages of cross-pollination.

Solution:

Cross-pollination is the transfer of pollen grains from the anther of one flower to the stigma of another flower of the same species. It is mainly brought about by agents such as wind, water, insects, and animals.

Advantages of cross-pollination:

1. **Genetic variation:** Produces new combinations of traits, increasing adaptability.
2. **Eliminates defective traits:** Recessive harmful mutations are masked by dominant healthy alleles.
3. **Hybrid vigour:** Offspring are more vigorous, disease-resistant, and better yielding.
4. **Evolutionary significance:** Increases diversity, which drives evolution of new varieties.

Conclusion.

Cross-pollination ensures survival, adaptability, and improvement of plant species.

Quick Tip

Cross-pollination = more variation, better adaptability, hybrid vigour. Self-pollination = purity of traits but less variation.

(e) What is biodiversity hotspot?

Solution:

A **biodiversity hotspot** is a biogeographic region that is rich in endemic species and has a high level of biodiversity, but is under severe threat from human activities. The concept was introduced by Norman Myers in 1988.

Criteria for a Region to Qualify as a Hotspot:

1. Must have at least **1,500 species of vascular plants as endemics**.
2. Must have lost at least **70% of its original habitat**.

Examples of Biodiversity Hotspots in India:

- Indo-Burma region (including North-East India and Andaman-Nicobar Islands).
- Himalaya (entire Indian Himalayan region and Indo-Burma ranges).
- Indo-Malayan region (Nicobar Islands).
- Sundalands including Nicobar group of islands.

Significance:

- Hotspots cover only about 2.3% of Earth's land area but support more than 50% of endemic plant species.
- They are crucial for ecological balance and conservation of global biodiversity.

> Quick Tip
>
> Biodiversity hotspots = small areas, very rich in species (especially endemics), but highly threatened. Protecting them ensures global biodiversity conservation.

3.(a) Pollen grains get their nourishment from which part of anther?

Solution:
Answer: Pollen grains get their nourishment from the **tapetum**, the innermost layer of the anther wall.
Explanation: - The anther wall consists of four layers: epidermis, endothecium, middle layers, and tapetum. - The **tapetum** is a nutritive tissue rich in cytoplasm and organelles. - It provides enzymes, hormones, amino acids, and other nutrients required for the development and maturation of pollen grains. - Tapetal cells also produce **sporopollenin**, a complex substance that forms the tough outer wall (exine) of pollen grains, making them resistant to environmental stress.
Conclusion: Thus, the tapetum plays a crucial role in nourishing developing pollen grains and ensuring successful male gametophyte formation.

> Quick Tip
>
> Remember: Tapetum = nutritive tissue of anther → supplies food + sporopollenin for pollen wall.

(b) Comment upon amniocentesis.

Solution:
Definition: Amniocentesis is a prenatal diagnostic technique in which a small amount of amniotic fluid is withdrawn from the uterus of a pregnant woman using a hollow needle, under ultrasound guidance.
Purpose of Amniocentesis:

- To detect chromosomal abnormalities (e.g., Down syndrome, Turner syndrome).
- To diagnose genetic disorders (e.g., sickle cell anemia, cystic fibrosis).
- To determine fetal sex and detect sex-linked disorders.

- To study lung maturity in late pregnancy.

Misuse of Amniocentesis: - In some countries, including India, amniocentesis was misused for **female feticide** by determining the sex of the fetus. - This led to a decline in the female child ratio.
Legal Aspect in India: - The **Pre-Conception and Pre-Natal Diagnostic Techniques (PCPNDT) Act, 1994** strictly prohibits the misuse of amniocentesis for sex determination. - It can only be used for detecting genetic and chromosomal abnormalities.
Conclusion: Amniocentesis is a valuable medical tool when used ethically for detecting genetic diseases, but its misuse for sex determination is illegal and punishable.

Quick Tip

Amniocentesis = prenatal diagnostic test for genetic diseases, not for sex determination (illegal in India under PCPNDT Act).

(c) Differentiate between true fruit and false fruit.

Solution:

Feature	**True Fruit**	**False Fruit**
Origin	Develops only from the ovary after fertilization.	Develops from ovary along with other floral parts such as thalamus, receptacle, or sepals.
Example	Mango, Tomato, Banana.	Apple, Strawberry, Cashew.
Structure	Pericarp differentiates into epicarp, mesocarp, and endocarp.	Edible part often arises from thalamus or other accessory floral structures.

Conclusion.
Thus, a true fruit arises solely from the ovary, while a false fruit includes additional floral parts.

Quick Tip

True fruit = ovary only. False fruit = ovary + other floral parts (e.g., apple from thalamus).

(d) Define Apomixis.

Solution:

Step 1: Definition.
Apomixis is a form of asexual reproduction that mimics sexual reproduction, where seeds are formed without fertilization.
Step 2: Characteristics.

- Seeds are produced without the union of gametes.
- Offspring are genetically identical to the parent (clones).
- Helps maintain hybrid vigour for several generations.

Step 3: Examples.

- Citrus, Mango, Grass species.

Step 4: Importance.
Apomixis is important in plant breeding because it helps fix desirable traits and maintain superior hybrids.

> Quick Tip
>
> Apomixis = seed formation without fertilization, producing clones (e.g., Citrus, Mango).

(e) What do you mean by Polyembryony?

Solution:
Polyembryony is the phenomenon of formation of more than one embryo from a single fertilized egg (zygote) or from other structures of the ovule in plants and animals.
Types of Polyembryony:

1. **True Polyembryony:** Multiple embryos are formed from a single fertilized egg. Example: Identical twins in humans.
2. **False Polyembryony:** Additional embryos are formed from nucellus or integuments of the ovule. Example: Citrus and Mango show polyembryony due to adventive embryos from nucellar cells.

Significance of Polyembryony:

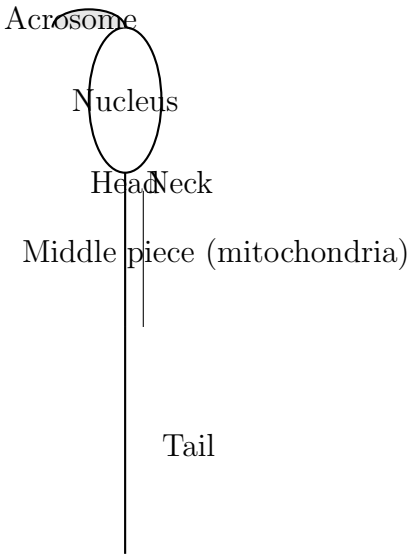
- Ensures survival advantage by producing more than one embryo.
- Helps in horticulture and plant breeding as nucellar embryos are genetically identical to the mother plant.
- Provides a natural method of cloning in some plants.

Quick Tip

Polyembryony = multiple embryos from a single seed. Examples: Citrus, Mango (plants); Identical twins (animals).

4.(a) Draw a labelled diagram of a human sperm.

Solution:
A human sperm is a haploid male gamete responsible for fertilization. It has four main parts: head, neck, middle piece, and tail.



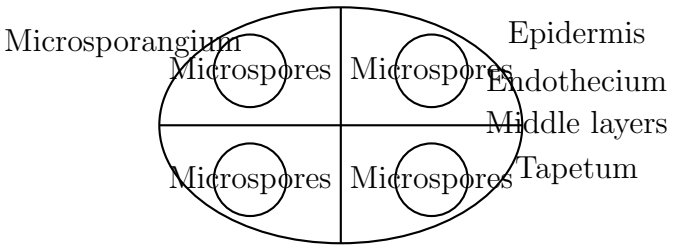
Explanation:
- **Head:** Contains nucleus with haploid chromosomes; acrosome has enzymes to digest egg coverings.
- **Neck:** Connects head and middle piece.
- **Middle piece:** Rich in mitochondria for energy (ATP) to power movement.
- **Tail:** Provides motility to reach the egg.

Quick Tip

Sperm = Head (nucleus + acrosome) + Middle piece (mitochondria) + Tail (movement).

(b) Draw a labelled diagram of a transverse section of a mature anther.

Solution:
A mature anther is bilobed and each lobe has two microsporangia, making it tetrasporangiate.



Explanation: Layers of anther wall (from outside to inside):

1. **Epidermis:** Protective outer layer.
2. **Endothecium:** Provides mechanical support.
3. **Middle layers:** 2–4 layers that degenerate at maturity.
4. **Tapetum:** Innermost nutritive layer providing food and enzymes to developing pollen grains.

Each microsporangium produces pollen grains after meiosis of microspore mother cells.

> Quick Tip
>
> Anther is tetrasporangiate: 2 lobes $\times$ 2 microsporangia each = 4 pollen sacs.

(c) What do you mean by bee-keeping? Explain it in brief.

Solution:

Step 1: Definition.
Bee-keeping, also called **apiculture**, is the scientific method of rearing and management of honeybees for the production of honey, beeswax, and other products.
Step 2: Requirements for bee-keeping.

- Knowledge of the biology and behaviour of honeybees.
- Availability of suitable location for bee colonies.
- Selection of suitable species like *Apis mellifera* (Italian bee) or *Apis cerana indica* (Indian bee).
- Use of bee-hives, proper maintenance, and protection from diseases and predators.

Step 3: Importance of bee-keeping.

- Provides honey (nutritious food and medicine).
- Produces beeswax (used in cosmetics, polishes, and pharmaceuticals).
- Helps in pollination, increasing crop yield.

- Generates rural employment and supplementary income.

Conclusion.
Bee-keeping is an important agro-based industry with economic as well as ecological benefits.

> Quick Tip
>
> Bee-keeping = apiculture. It gives honey, wax, pollination benefits, and rural income.

(d) Write a short note on Family Planning.

Solution:

Step 1: Definition.
Family planning refers to practices adopted by individuals or couples to regulate the number and spacing of children according to their health, financial, and social conditions.
Step 2: Objectives.

- To control population growth.
- To ensure maternal and child health.
- To improve quality of life by preventing unwanted pregnancies.

Step 3: Methods of family planning.

1. **Natural methods:** Rhythm method, coitus interruptus, lactational amenorrhea.
2. **Barrier methods:** Condoms, diaphragms.
3. **Hormonal methods:** Oral pills, injectables, implants.
4. **Intrauterine devices (IUDs):** Copper-T, multiload devices.
5. **Surgical methods:** Vasectomy (male), tubectomy (female).

Step 4: Benefits.
Family planning helps in controlling population explosion, reduces poverty, improves women's health, and promotes socio-economic development.

> Quick Tip
>
> Family planning = controlling birth through natural, barrier, hormonal, IUD, or surgical methods to ensure health and balanced population growth.

5.(a) Give an account on DNA fingerprinting.

Solution:
DNA fingerprinting is a technique used to identify individuals based on their unique DNA sequence patterns. It was developed by Alec Jeffreys in 1984. In India, it was pioneered by Dr. Lalji Singh, known as the "Father of DNA Fingerprinting in India."
Principle: It is based on the polymorphism in repetitive DNA sequences, particularly **VNTRs (Variable Number of Tandem Repeats)** that vary from individual to individual.
Steps of DNA Fingerprinting:

1. **DNA Isolation:** Extraction of DNA from cells such as blood, hair root, or saliva.
2. **Restriction Digestion:** DNA is cut into fragments using restriction endonucleases.
3. **Separation:** Fragments are separated by gel electrophoresis.
4. **Hybridization:** Fragments are transferred to a nylon membrane and hybridized with radioactive DNA probes specific to VNTRs.
5. **Autoradiography:** The pattern of bands obtained on X-ray film represents the DNA fingerprint.

Applications:

- Criminal investigations and forensic science.
- Paternity and maternity testing.
- Identification of genetic disorders.
- Study of genetic diversity and evolutionary biology.

Quick Tip

DNA fingerprinting = "genetic barcode" unique to every person, useful in forensics, paternity testing, and genetic studies.

(b) Who is father of Green Revolution in India? Green Revolution is useful for farmers. Explain it with example.

Solution:
Father of Green Revolution in India: Dr. M.S. Swaminathan is regarded as the "Father of Green Revolution in India."
Green Revolution: It refers to the period (1960s–1970s) when high-yielding varieties (HYVs) of crops, modern irrigation techniques, chemical fertilizers, and pesticides were introduced to increase agricultural productivity in India.
Usefulness for Farmers:

- Increased food grain production, especially wheat and rice.
- Reduced dependency on food imports and ensured self-sufficiency.
- Enhanced rural employment through higher crop intensity.
- Improved economic condition of farmers due to surplus production.

Example: Introduction of HYV wheat varieties such as *Lerma Rojo* and *Sonora 64* in Punjab and Haryana drastically increased wheat production, making India self-sufficient in food grains.

Quick Tip

Green Revolution = HYV seeds + irrigation + fertilizers + pesticides. Increased wheat and rice yields, ensuring food security in India.

(c) What do you mean by restriction enzymes? Explain with example.

Solution:
Definition: Restriction enzymes, also called **restriction endonucleases**, are enzymes that cut DNA molecules at specific recognition sequences (usually palindromic sequences). They are also known as **molecular scissors**.
Characteristics:

- They recognize short, specific DNA sequences (4–8 base pairs).
- They cut DNA at specific sites, producing either **blunt ends** or **sticky ends**.
- They were first discovered in bacteria, where they protect against invading viral DNA.

Example: - **EcoRI:** Recognizes the sequence `GAATTC` and cuts between G and A, producing sticky ends. - **HindIII:** Recognizes the sequence `AAGCTT` and cuts to produce sticky ends.
Diagram (EcoRI Action):

$$\begin{aligned} &\text{5' - GAATTC - 3'} \xrightarrow{\text{EcoRI}} \text{5' - G AATTC - 3'} \\ &\text{3' - CTTAAG - 5'} \xrightarrow{\text{EcoRI}} \text{3' - CTTAA G - 5'} \end{aligned}$$

Importance: Restriction enzymes are widely used in recombinant DNA technology to cut and insert genes into vectors for genetic engineering.

Quick Tip

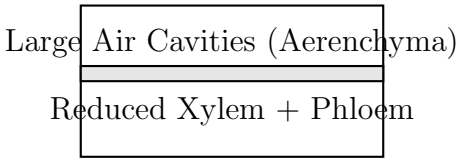
Restriction enzymes = molecular scissors. EcoRI and HindIII are commonly used in DNA technology.

(d) Hydrophytic plants are adapted to their environment. Explain it with the help of examples and diagrams.

Solution:
Definition: Hydrophytes are plants that grow in water or in very wet conditions. They show special morphological and anatomical adaptations to survive in such environments.
Adaptations of Hydrophytic Plants:

1. **Morphological Adaptations:** - Leaves are thin, large, and broad for floating (e.g., Lotus). - Some plants have dissected leaves to resist water currents (e.g., Hydrilla). - Stomata are found only on the upper surface of floating leaves (e.g., Water lily).
2. **Anatomical Adaptations:** - Presence of large **aerenchyma** (air spaces) for buoyancy. - Reduced xylem (since water transport is not required much). - Poorly developed mechanical tissues due to absence of mechanical stress.
3. **Reproductive Adaptations:** - Flowers are often large and attractive for pollination (e.g., Lotus). - In submerged plants, pollen may be transferred by water (hydrophily).

Examples: - *Lotus, Eichhornia, Hydrilla, Vallisneria.*
Diagram (Structure of Hydrophytic Leaf with Aerenchyma):



Leaf Section of Hydrophyte

Conclusion: Hydrophytes possess structural and physiological adaptations like aerenchyma, reduced mechanical tissues, and modified leaves that help them survive and reproduce successfully in aquatic environments.

Quick Tip

Hydrophytes show buoyancy (aerenchyma), reduced xylem, and floating leaves. Example: Lotus, Eichhornia.

6.(a) Name the important components of biodiversity and mention suitable measures for its conservation.

Solution:

Components of Biodiversity: Biodiversity can be studied at three major levels:

1. **Genetic Diversity:** Variation of genes within a species. - Example: Different rice varieties or breeds of dogs.

2. **Species Diversity:** Variety of species within a region. - Example: Western Ghats of India have a high diversity of amphibian species.
3. **Ecosystem Diversity:** Variety of ecosystems in a geographical area. - Example: Deserts, rainforests, wetlands, coral reefs, etc.

Measures for Conservation of Biodiversity:

- **In-situ Conservation:** - Protecting species in their natural habitat. - Examples: National Parks, Wildlife Sanctuaries, Biosphere Reserves, Sacred Groves.
- **Ex-situ Conservation:** - Conserving species outside their natural habitat. - Examples: Botanical Gardens, Zoos, Seed Banks, Cryopreservation.
- **Legal and Social Measures:** - Enforcing Wildlife Protection Act (1972), CITES, and Biodiversity Act. - Raising awareness through education and community participation.

Conclusion: Conservation of biodiversity is crucial for ecological balance, sustainable development, and survival of future generations.

Quick Tip

Biodiversity = Genetic + Species + Ecosystem diversity. Conservation = In-situ (parks) + Ex-situ (zoos, seed banks).

(b) Write short note on: (i) Genetical disorder in man (ii) Central dogma

Solution:

(i) Genetical Disorder in Man: Genetic disorders are abnormalities caused due to defects in genes or chromosomes. They are of two main types:

- **Mendelian Disorders (Single-gene defects):** - Inherited in predictable Mendelian patterns. - Examples: Sickle Cell Anemia (defective hemoglobin gene), Cystic Fibrosis, Phenylketonuria.
- **Chromosomal Disorders:** - Caused by numerical or structural abnormalities of chromosomes. - Examples: Down's Syndrome (Trisomy-21), Turner's Syndrome (Monosomy of X chromosome), Klinefelter's Syndrome (XXY condition).

Conclusion: Genetic disorders are mostly incurable but can be managed with genetic counseling, prenatal diagnosis, and supportive treatments.

(ii) Central Dogma: - Proposed by Francis Crick in 1958. - It describes the unidirectional flow of genetic information in all living organisms:

$$\text{DNA} \xrightarrow{\text{Transcription}} \text{RNA} \xrightarrow{\text{Translation}} \text{Protein}$$

- **Explanation:** - DNA serves as a template to synthesize RNA (transcription). - RNA carries information to ribosomes where proteins are synthesized (translation). - Proteins then control cellular structure and function.
Conclusion: The central dogma explains the fundamental process of molecular biology and highlights how genetic information is expressed in cells.

Quick Tip

Genetic disorders = Mendelian (Sickle cell) + Chromosomal (Down's syndrome). Central dogma = DNA $\rightarrow$ RNA $\rightarrow$ Protein.

(c) What is bio-fertilizer? Describe its role in the production of food.

Solution:

Step 1: Definition.
Bio-fertilizers are living microorganisms which, when applied to seeds, plant surfaces, or soil, promote growth by increasing the availability of essential nutrients to plants.
Step 2: Examples of bio-fertilizers.

- **Rhizobium:** Symbiotic bacteria in root nodules of legumes; fix atmospheric nitrogen.
- **Azotobacter and Azospirillum:** Free-living nitrogen-fixing bacteria.
- **Cyanobacteria (Blue-Green Algae like Anabaena, Nostoc):** Fix nitrogen in paddy fields.
- **Mycorrhizal fungi:** Enhance phosphorus and water absorption.

Step 3: Role in food production.

- Increase soil fertility naturally without chemical fertilizers.
- Promote sustainable agriculture by reducing pollution.
- Improve crop yield, especially in cereals, legumes, and paddy.
- Cost-effective and eco-friendly compared to synthetic fertilizers.

Conclusion.
Bio-fertilizers are essential for eco-friendly, sustainable agriculture and play a vital role in ensuring long-term food security.

Quick Tip

Bio-fertilizers = microbes that enrich soil nutrients (e.g., Rhizobium, Azotobacter, Cyanobacteria). They reduce chemical fertilizer use.

(d) Write short note on any two:
(i) Red data book
(ii) Parasitism
(iii) Role of leguminous plants in increasing the fertility of soil.

Solution:

(i) Red data book:

- The Red Data Book is a record of endangered and threatened species of plants and animals.
- It is maintained by the International Union for Conservation of Nature (IUCN).
- It helps in monitoring species at risk of extinction and guiding conservation strategies.

(ii) Parasitism:

- A type of symbiotic relationship where one organism (parasite) derives nutrition from another organism (host), causing harm to the host.
- Examples: Plasmodium (malarial parasite) in humans, Cuscuta (dodder) on other plants.

(iii) Role of leguminous plants in increasing soil fertility:

- Leguminous plants like peas, beans, and clover have root nodules containing **Rhizobium** bacteria.
- These bacteria fix atmospheric nitrogen into nitrates, enriching the soil.
- Reduces the need for synthetic nitrogen fertilizers and improves crop productivity.

Quick Tip

Red Data Book = list of threatened species. Parasitism = one benefits, host harmed. Legumes = nitrogen fixation improves soil fertility.

---

7. What is Megasporogenesis? Describe the development of zygote up to formation of mature ovule in an Angiospermic flower.

Solution:

Megasporogenesis: It is the process of formation of haploid megaspores from a diploid megaspore mother cell (MMC) in the ovule of flowering plants (angiosperms).
Step 1: Formation of Megaspore Mother Cell (MMC).

- In the ovule, a single hypodermal cell differentiates into an MMC (2n).
- The MMC is large with dense cytoplasm and a prominent nucleus.

Step 2: Meiosis in MMC.

- The MMC undergoes meiosis to form a linear tetrad of four haploid megaspores.
- Usually, three megaspores degenerate and one functional megaspore survives.

Step 3: Development of Female Gametophyte (Embryo Sac).

- The functional megaspore undergoes three mitotic divisions → 8 nuclei.
- Arrangement: 3 cells at micropylar end (egg apparatus: 1 egg + 2 synergids), 3 cells at chalazal end (antipodals), and 2 polar nuclei in the center (fuse to form secondary nucleus).

Final Structure: The mature ovule contains a 7-celled, 8-nucleate female gametophyte (embryo sac).

Quick Tip

Megasporogenesis = MMC (2n) → meiosis → 4 megaspores → 1 functional megaspore → embryo sac (7-celled, 8-nucleate).

OR
Draw a labeled diagram of reproductive system of a woman.

Solution:

The female reproductive system consists of a pair of ovaries, fallopian tubes, uterus, cervix, and vagina.
Key Structures:

- **Ovaries:** Produce eggs (ova) and hormones (estrogen, progesterone).
- **Fallopian Tubes (Oviducts):** Carry the ovum; fertilization occurs in the ampullary region.
- **Uterus:** Muscular organ where embryo implants and develops.
- **Cervix:** Lower part of uterus, opens into vagina.
- **Vagina:** Birth canal and site of copulation.

Diagram (to be drawn in exam): A neat, labeled diagram should include:

- Ovaries
- Fallopian tubes (fimbriae, infundibulum, ampulla, isthmus)
- Uterus

- Cervix
- Vagina

> Quick Tip
>
> Always label ovaries, fallopian tubes, uterus, cervix, and vagina clearly when drawing the female reproductive system.

---

8. Write short note on any two: (i) Double fertilization (ii) Sewage treatment (iii) Control of Pneumonia

Solution:

(i) Double Fertilization: - It is a unique feature of flowering plants (angiosperms). - When a pollen tube enters the embryo sac, it releases two male gametes.

- One male gamete fuses with the egg cell to form a diploid zygote (2n) → **Syngamy**.
- The other male gamete fuses with two polar nuclei to form a triploid (3n) endosperm nucleus → **Triple fusion**.

- Since two types of fertilization events occur (syngamy + triple fusion), it is called **double fertilization**. - Significance: Endosperm formed provides nourishment to the developing embryo.

(ii) Sewage Treatment: - Sewage treatment is a process of removing contaminants from wastewater before releasing it into water bodies. It has two main steps:

1. **Primary Treatment:** - Physical removal of large and suspended particles using sedimentation, filtration, and skimming.
2. **Secondary Treatment:** - Biological treatment using aerobic microbes in aeration tanks, which degrade organic matter. - Sludge formed is anaerobically digested in biogas plants to produce methane.

- Importance: Prevents water pollution, protects aquatic life, and recycles waste into biogas and manure.

(iii) Control of Pneumonia: - Pneumonia is a serious disease caused by bacteria like *Streptococcus pneumoniae* or fungi/viruses. - **Symptoms:** Inflammation of alveoli, fever, cough, chest pain, and difficulty in breathing. - **Control and Prevention:**

- Vaccination against pneumococcus.
- Antibiotics (penicillin, erythromycin) for bacterial pneumonia.
- Good nutrition, hygiene, and avoiding overcrowded places reduce infection chances.

Quick Tip

Double fertilization is unique to angiosperms. Sewage treatment = Primary + Secondary stages. Pneumonia can be prevented by vaccination and treated with antibiotics.

OR
Differentiate between the following fruits:
(i) Legume and Lomentum
(ii) Drupe and Schizocarpic fruit

Solution:

(i) Legume vs Lomentum:

Legume	**Lomentum**
Simple, dry, dehiscent fruit that splits along both sutures.	Modified legume that breaks into one-seeded segments (indehiscent).
Example: Pea, Bean.	Example: Acacia, Mimosa.

(ii) Drupe vs Schizocarpic Fruit:

Drupe	**Schizocarpic Fruit**
A simple, fleshy fruit with three distinct layers: epicarp, mesocarp (fleshy), and endocarp (hard, stony).	A dry, many-seeded fruit that splits into separate one-seeded indehiscent parts (mericarps).
Example: Mango, Coconut.	Example: Coriander, Cotton.

Quick Tip

Legume splits along both sutures; Lomentum breaks into one-seeded segments. Drupe = fleshy with stony endocarp; Schizocarpic = dry fruit splitting into mericarps.

9.(a) Write short note on:
(i) Fertilization in human
(ii) Active and Passive Immunity

Solution:

(i) Fertilization in human:

- Fertilization is the process of fusion of the male gamete (sperm) and the female gamete (ovum) to form a diploid zygote.

- It usually occurs in the **ampulla of the fallopian tube**.
- Steps:
  1. **Capacitation:** Sperm undergo physiological changes in the female tract to penetrate the ovum.
  2. **Acrosomal reaction:** Enzymes from the acrosome help the sperm penetrate zona pellucida of the ovum.
  3. **Fusion:** Plasma membranes of sperm and ovum fuse, and the sperm nucleus enters the cytoplasm.
  4. **Cortical reaction:** Prevents polyspermy (entry of additional sperms).
  5. **Zygote formation:** Male and female nuclei fuse to form the diploid zygote (2n = 46).
- **Significance:** Restores diploid chromosome number, determines sex of offspring, and initiates embryonic development.

(ii) Active and Passive Immunity:

- **Active Immunity:**
  - Achieved when the body produces its own antibodies against antigens.
  - Can be natural (after infection) or artificial (after vaccination).
  - Provides long-lasting protection due to memory cells.
  - Example: Immunity after polio vaccination.
- **Passive Immunity:**
  - Ready-made antibodies are introduced into the body.
  - Provides immediate but short-term protection.
  - Example: Anti-tetanus serum, mother-to-child antibodies via placenta or colostrum.

Conclusion.
Active immunity is long-lasting and involves memory cells, whereas passive immunity provides immediate but temporary protection.

Quick Tip

Fertilization = sperm + ovum → zygote in fallopian tube. Active immunity = body makes antibodies, long-term. Passive = ready-made antibodies, short-term.

OR

(b) Write an essay on Genetic code.

Solution:

Step 1: Definition.
The genetic code is the set of rules by which the sequence of nucleotides in DNA or mRNA is translated into the sequence of amino acids in proteins. Codons (triplets of nucleotides) specify particular amino acids.
Step 2: Properties of the genetic code.

1. **Triplet code:** Each codon has 3 nucleotides.
2. **Unambiguous:** One codon codes for only one amino acid.
3. **Degenerate:** A single amino acid may be coded by more than one codon (e.g., leucine).
4. **Universal:** Same code is used by nearly all organisms (exceptions in mitochondria).
5. **Comma-less:** Codons are read continuously without gaps.
6. **Start codon:** AUG (codes for methionine) initiates translation.
7. **Stop codons:** UAA, UAG, UGA terminate translation.

Step 3: Importance.

- Ensures accurate protein synthesis.
- Universal nature supports the theory of common ancestry.
- Degeneracy reduces the effect of mutations (silent mutations possible).

Step 4: Conclusion.
The genetic code is a precise, universal language of life that ensures transfer of information from DNA to proteins, sustaining all forms of life.

Quick Tip

Genetic code = triplet, universal, degenerate, unambiguous. Start = AUG, Stop = UAA, UAG, UGA.
